Temperature sensor

DETECT | RASPBERRY PI | INPUT | ANALOGUE

What does it do?

A TMP36 turns temperature into a changing voltage that a program can measure.

Its output rises by 10 millivolts for every degree Celsius, with 0.5 V representing 0 °C.

The Pico reads that voltage directly; the Raspberry Pi uses an MCP3008 analogue-to-digital converter.

Uses



Refrigerator cooling

DETECT Measure fridge temperature → **DECIDE** Is it too warm? → **DO** Increase the cooling



Soup drama meter

DETECT Measure bowl surface → **DECIDE** Which drama level fits? → **DO** Move the drama pointer



Flat face	+Vs	Vout	GND
Use it to identify the pins	Left pin	Centre pin	Right pin

Raspberry Pi Foundation

Code

Raspberry Pi

```
from gpiozero import MCP3008

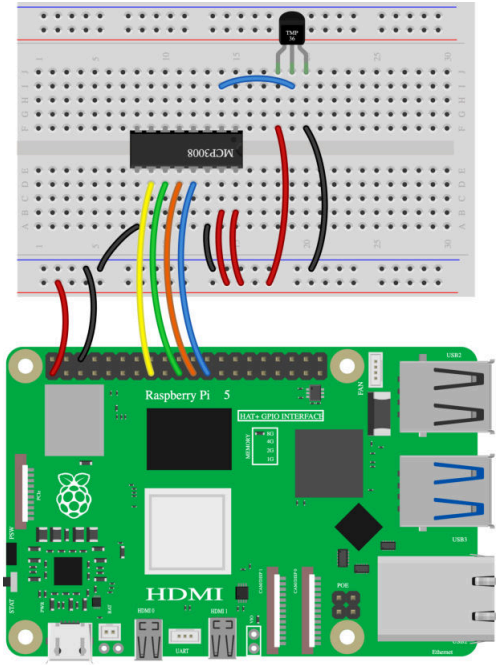
sensor = MCP3008(channel=1, select_pin=22)

temperature = (sensor.voltage - 0.5) * 100
```

Read the TMP36 with its flat face towards you; the pins are +Vs, Vout, GND from left to right.

Wiring

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TMP36 **Vout** → MCP3008 **CH1**. CLK / DIN / DOUT / CS → **GPIO 11 / 10 / 9 / 22** (pins 23 / 19 / 21 / 15). **3V3** (pin 1), **GND** (pin 6).